

R E S E A R C H P R O J E C T

Impact of Inflation and Interest Rates on Indian Stock Market Returns

An Empirical Analysis of NIFTY 50, CPI Inflation, and RBI Repo Rate

Sample Period: January 2014 – November 2023 | 119 Monthly Observations

Data Sources: NSE India, Reserve Bank of India, Ministry of Statistics and Programme Implementation

Abstract

This research project examines the relationship between macroeconomic variables and short-term stock market returns in India over the period January 2014 to November 2023. Using monthly data on the NIFTY 50 index, Consumer Price Index (CPI) based inflation, and the Reserve Bank of India (RBI) Repo Rate, an Ordinary Least Squares (OLS) regression framework is employed to quantify the impact of inflation and monetary policy on equity market performance.

The empirical results reveal a statistically weak relationship between contemporaneous inflation, repo rate, and NIFTY 50 monthly returns, with an R-squared of approximately 0.54% and both coefficients insignificant at conventional confidence levels. These findings are consistent with the efficient markets hypothesis, which posits that publicly anticipated macroeconomic information is rapidly incorporated into asset prices. Additional analysis using lagged inflation, real interest rates, and inflation regime classification provides supplementary insights into delayed transmission effects and the differential behaviour of markets under high and normal inflation environments.

The study contributes to the growing body of literature on macroeconomic determinants of emerging market equity returns and highlights the limitations of contemporaneous single-factor models in explaining short-run stock price movements.

Keywords: *NIFTY 50, Inflation, Repo Rate, CPI, OLS Regression, Efficient Markets, Emerging Markets, Real Interest Rate, Inflation Regime*

1. Introduction

The relationship between macroeconomic variables and stock market performance has been a subject of extensive inquiry in financial economics. In an emerging economy such as India, where capital markets are rapidly deepening and monetary policy is an active instrument of economic management, understanding how inflation and interest rates influence equity returns carries significant practical and academic importance.

India's equity markets have witnessed remarkable growth over the past decade. The NIFTY 50 index, which tracks the performance of the fifty largest companies listed on the National Stock Exchange (NSE), rose from approximately 6,000 points in early 2014 to nearly 20,000 points by late 2023. Over the same period, the Indian economy navigated several macroeconomic cycles, including episodes of elevated inflation, aggressive monetary tightening, a global pandemic, and subsequent recovery. These structural shifts provide a rich empirical environment for studying the interplay between macroeconomic conditions and financial markets.

From a theoretical standpoint, higher inflation erodes the real value of corporate earnings and raises input costs, potentially depressing stock valuations. Meanwhile, rising interest rates increase the discount rate applied to future cash flows, which mechanically reduces the present value of equities. At the same time, financial markets are forward-looking institutions. Investors do not simply react to macroeconomic data as it is released; rather, they form expectations and position their portfolios in advance. Consequently, by the time official data on inflation or policy rates becomes publicly available, it may already be fully reflected in market prices.

This project seeks to test empirically whether short-term NIFTY 50 returns exhibit a statistically meaningful dependence on contemporaneous and lagged inflation and repo rate changes over the 2014–2023 sample period.

1.1 Research Objectives

The primary objectives of this study are:

- To measure and model the relationship between CPI-based inflation, the RBI Repo Rate, and monthly NIFTY 50 returns using OLS regression.
- To examine whether lagged inflation provides greater explanatory power than contemporaneous inflation.
- To construct a real interest rate series and assess its conceptual significance for monetary policy analysis.
- To compare average NIFTY 50 returns across high-inflation and normal-inflation regimes as defined by the RBI's tolerance band.

- To interpret the empirical findings in light of established theories of market efficiency and macroeconomic transmission.

1.2 Significance of the Study

India presents a uniquely interesting case for this investigation. As one of the world's fastest-growing major economies, India's monetary policy framework underwent a landmark transformation in 2016 with the adoption of a flexible inflation targeting regime, anchored at 4% with a tolerance band of $\pm 2\%$. The Repo Rate, therefore, is not merely a price signal but an explicit instrument of inflation management. Examining how markets respond to this policy variable—and to the inflation it targets—is of direct relevance to investors, policymakers, and researchers alike.

2. Literature Review

The academic literature on the macroeconomic determinants of stock returns is substantial. Early theoretical foundations were laid by the Capital Asset Pricing Model (CAPM) and the Arbitrage Pricing Theory (APT), the latter of which explicitly accounts for macroeconomic risk factors as drivers of asset returns.

Fama and Schwert (1977) were among the first to examine systematically the relationship between stock returns and inflation, finding a negative relationship in US data—a finding that challenged the long-held view that equities serve as a hedge against inflation. Subsequent studies in developed markets largely corroborated this negative association, though its magnitude and significance were found to vary with the time horizon and the economic regime.

In the context of emerging markets, the evidence is more nuanced. Studies focusing on South Asian and Southeast Asian economies have generally found weaker and often statistically insignificant contemporaneous relationships between inflation, interest rates, and equity returns. This is attributable partly to higher market volatility in these economies and partly to the role of global capital flows, which can dominate domestic macroeconomic signals in driving short-run price movements.

For India specifically, a strand of literature using monthly and quarterly data over various sample periods reports mixed evidence. Some studies document a significant negative relationship between interest rate changes and stock returns, consistent with the standard discounted cash flow logic. Others find that the relationship breaks down or reverses during periods of high growth, where investors prioritise economic momentum over policy tightening signals. The efficient markets hypothesis (Fama, 1970) provides a reconciling framework: if macroeconomic information is rapidly and fully incorporated into prices, contemporaneous regressions will yield insignificant coefficients regardless of the underlying economic relationship.

More recent work has emphasised the importance of inflation regimes and the role of the real interest rate. Real rates—defined as the nominal policy rate less the prevailing inflation rate—capture the effective stance of monetary policy in a way that nominal rates alone cannot. Negative real rates, which signal loose monetary conditions, have been associated with elevated asset price inflation across multiple economies.

This project draws on and extends this literature by applying a rigorous data construction process, employing multiple analytical lenses—regression, correlation, regime analysis—and providing a clear, evidence-based interpretation of the findings in the Indian context.

3. Data and Methodology

3.1 Data Sources and Sample Period

The empirical analysis employs monthly data spanning January 2013 to November 2023. However, owing to the construction of the year-on-year inflation measure—which requires twelve prior months of CPI data—the effective regression sample begins in January 2014 and contains 119 monthly observations. Three primary data series were collected:

- NIFTY 50 Index: Monthly closing values from the National Stock Exchange of India (NSE).
- Consumer Price Index (CPI): Combined (rural + urban) CPI from the Ministry of Statistics and Programme Implementation (MoSPI). The combined CPI is used as it provides the most comprehensive measure of economy-wide price changes.
- Repo Rate: The policy interest rate set by the Reserve Bank of India (RBI), sourced from RBI publications. The repo rate is maintained constant between policy meetings and updated in the dataset at the date of each announced policy change.

3.2 Variable Construction

Four key analytical variables were constructed from the raw data:

NIFTY 50 Monthly Returns

Raw index levels were transformed into monthly percentage returns using the standard formula:

$$Return_t = [(P_t - P_{t-1}) / P_{t-1}] \times 100$$

where P_t denotes the closing index value in month t . The use of returns rather than raw price levels eliminates the non-stationarity inherent in trending price series, a prerequisite for valid OLS inference.

Year-on-Year CPI Inflation

Inflation was measured on a year-on-year (YoY) basis to remove seasonal distortions:

$$Inflation_t = [(CPI_t - CPI_{t-12}) / CPI_{t-12}] \times 100$$

This approach compares each month's price level to the same month in the previous year, neutralising predictable seasonal patterns in consumer spending. Where the raw CPI dataset contained missing values, linear interpolation was applied, a methodology justified by the gradual, smooth nature of price index movements.

Lagged Inflation

A one-month lag of the inflation series was constructed to capture potential delayed market responses. Since official inflation data is published with a lag, investors may react to the prior month's released figure rather than the current month's unannounced figure.

Real Interest Rate

The real interest rate was computed as:

$$Real\ Rate_t = Repo\ Rate_t - Inflation_t$$

This measure captures the effective monetary policy stance, accounting for the erosive effect of inflation on the nominal rate's real purchasing power implications. Negative real rates indicate accommodative monetary conditions.

3.3 Inflation Regime Classification

Months were classified into two inflation regimes based on the RBI's formal inflation targeting framework:

- High Inflation: Year-on-year CPI inflation exceeding 6%, representing a breach of the upper tolerance band of the RBI's 4% $\pm$ 2% target range.
- Normal Inflation: Year-on-year CPI inflation at or below 6%, i.e., within the policy tolerance band.

This classification enables a comparison of equity market behaviour across macroeconomic environments and allows us to assess whether the weak aggregate results mask heterogeneity across regimes.

3.4 Regression Model

The primary analytical tool is an Ordinary Least Squares (OLS) multiple regression model. The baseline specification is:

$$Return_t = \beta_0 + \beta_1 \times Inflation_t + \beta_2 \times RepoRate_t + \varepsilon_t$$

where $Return_t$ is the monthly NIFTY 50 return, $Inflation_t$ is year-on-year CPI inflation, $RepoRate_t$ is the RBI Repo Rate, and ε_t is the error term assumed to satisfy standard OLS assumptions. The coefficients β_1 and β_2 capture the marginal effects of inflation and the repo rate on stock returns, holding the other variable constant.

4. Descriptive Statistics

Before examining the regression results, it is instructive to characterise the distribution of each variable over the sample period. Table 1 presents the summary statistics for the primary variables.

Table 1: Summary Statistics (January 2014 – November 2023)

Variable	Mean	Std. Deviation	Minimum	Maximum
NIFTY 50 Returns (%)	0.99	4.72	-14.15	14.88
CPI Inflation (%)	5.15	1.73	1.46	8.61
Repo Rate (%)	6.26	0.81	4.00	8.00
Real Interest Rate (%)	1.11	1.81	-3.43	4.83

The average monthly NIFTY 50 return over the sample period is approximately 0.99%, reflecting the broad bull market trend observed in Indian equities during the 2014–2023 period. The standard deviation of 4.72 percentage points indicates substantial month-to-month variability, with returns ranging from a low of roughly -14% (likely during the COVID-19 market shock of March 2020) to a high of approximately +15%.

Mean CPI inflation stands at approximately 5.15%, broadly within the RBI's tolerance band, though the range of 1.46% to 8.61% underscores the significant variation in price pressures over the decade. The repo rate averaged 6.26% over the sample, ranging from a pandemic-era low of 4.00% to a high of 8.00% in the tightening cycle of 2013–2014. The real interest rate averaged 1.11%, suggesting that monetary conditions were, on average, mildly restrictive in real terms.

4.1 Inflation Regime Distribution

Of the 119 monthly observations in the regression sample:

- 82 months (approximately 69%) were classified as Normal Inflation months ($\text{CPI} \leq 6\%$).
- 35 months (approximately 29%) were classified as High Inflation months ($\text{CPI} > 6\%$).

The regime distribution indicates that high inflation was not a rare occurrence during this period; it characterised nearly one-third of the sample. High inflation months were concentrated in 2013–2014 and again in 2022–2023, coinciding with global commodity price surges and post-pandemic supply-side pressures.

4.2 Correlation Analysis

A preliminary correlation analysis reveals a modest positive correlation of approximately 0.052 between inflation and NIFTY 50 returns. This sign is counterintuitive relative to standard theory, which predicts a negative relationship, but the magnitude is negligibly small and not statistically distinguishable from zero. The correlation between the repo rate and returns is negative at approximately -0.063, consistent with the standard theory that tighter monetary policy dampens equity performance, but again, this is economically and statistically negligible.

These preliminary results foreshadow the regression findings: the contemporaneous linear relationship between these macroeconomic variables and market returns is, at best, extremely weak.

5. Regression Results and Interpretation

5.1 Baseline Regression Output

Table 2 presents the full OLS regression results for the baseline model.

Table 2: OLS Regression Results – Baseline Model

Metric	Value
Multiple R	0.0733
R-Squared	0.0054
Adjusted R-Squared	-0.0118
Standard Error of Regression	4.7476
Number of Observations	119

Table 3: ANOVA Table

Source	df	SS	MS	F-Statistic	Significance F
Regression	2	14.13	7.07	0.3135	0.7315
Residual	116	2,614.61	22.54		
Total	118	2,628.74			

Table 4: Coefficient Estimates

Variable	Coefficient	Std. Error	t-Statistic	p-Value	95% CI Lower	95% CI Upper
Intercept (β_0)	1.7313	2.7368	0.6326	0.5282	-3.689	7.152

Inflation (β_1)	0.1234	0.2859	0.4317	0.6668	-0.443	0.690
Repo Rate (β_2)	-0.2139	0.3514	-0.6086	0.5440	-0.910	0.482

5.2 Interpretation of Results

The regression results yield several important insights:

Model Fit (R-Squared = 0.0054)

The R-squared value of approximately 0.54% implies that contemporaneous inflation and the repo rate together explain less than 1% of the variation in monthly NIFTY 50 returns. The adjusted R-squared is negative (-1.18%), indicating that the model performs worse than a simple mean-return benchmark. This is an extraordinarily low level of explanatory power and constitutes the central empirical finding of this study.

Inflation Coefficient ($\beta_1 = 0.1234$, $p = 0.667$)

The estimated coefficient on inflation is positive and small, suggesting that a one percentage point increase in CPI inflation is associated with an increase of approximately 0.12 percentage points in monthly NIFTY 50 returns, holding the repo rate constant. However, the p-value of 0.667 indicates that this coefficient is not statistically different from zero at any conventional significance level (1%, 5%, or 10%). The 95% confidence interval spans from -0.443 to +0.690, encompassing zero by a wide margin.

Repo Rate Coefficient ($\beta_2 = -0.2139$, $p = 0.544$)

The repo rate coefficient is negative, consistent with the theoretical expectation that higher borrowing costs reduce equity valuations. A one percentage point increase in the repo rate is associated with a decline of approximately 0.21 percentage points in monthly returns, ceteris paribus. However, as with the inflation coefficient, this estimate is statistically insignificant ($p = 0.544$), and the wide 95% confidence interval (-0.910 to +0.482) precludes any confident inference about the sign, let alone the magnitude.

F-Statistic and Joint Significance

The F-statistic of 0.3135 with a p-value of 0.7315 decisively fails to reject the null hypothesis that all slope coefficients are jointly zero. There is no statistically meaningful joint relationship between the two macroeconomic regressors and monthly equity returns in this sample.

5.3 Economic Interpretation

The absence of a statistically significant contemporaneous relationship is not necessarily indicative of a failure of economic theory. Several explanations are consistent with these findings:

- **Efficient Markets Hypothesis:** If Indian equity markets are informationally efficient, the macroeconomic information contained in the latest CPI release and the current repo rate should already be embedded in current prices. By the time these figures are measured and recorded in the dataset, markets have already reacted.
- **Forward-Looking Expectations:** Investors form expectations about future inflation and policy rates and adjust their portfolios accordingly. The realised monthly figures may add little incremental information beyond what markets had already anticipated.
- **Omitted Variables:** A simple bivariate model omits other critical drivers of equity returns, including global market conditions, corporate earnings surprises, exchange rate movements, and geopolitical events. These factors may dominate macroeconomic variables at the monthly frequency.
- **Non-Linear or Regime-Dependent Effects:** The relationship between inflation, interest rates, and returns may be non-linear or may only manifest strongly in extreme macroeconomic environments, which are diluted in an aggregate linear regression.

6. Extended Analysis

6.1 Lagged Inflation

To account for the possibility that markets react to officially released (and therefore lagged) inflation data rather than the current month's unannounced figure, a one-month lagged inflation variable was introduced into the analysis. Economic theory and empirical research in emerging markets suggest that data release lags can cause measured macroeconomic variables to affect prices with a delay.

In the present dataset, lagged inflation was constructed by shifting the inflation series one period forward. The preliminary assessment of lagged inflation's correlation with returns is similarly weak, consistent with the hypothesis that even publicly released prior-month inflation data is rapidly incorporated into prices in the current month. This finding is consistent with semi-strong form market efficiency, where all publicly available information—including last month's inflation figure—is already reflected in today's prices.

6.2 Real Interest Rate Analysis

The real interest rate provides a more economically meaningful measure of monetary policy stance than the nominal repo rate alone. Over the sample period, real interest rates ranged from a minimum of approximately -3.43% to a maximum of approximately +4.83%, with a mean of around +1.11%.

Periods of negative real interest rates, which occurred notably during the COVID-19 policy response of 2020–2021 when the repo rate was cut to 4% while inflation remained elevated, represent episodes

of highly accommodative monetary conditions. Theoretically, such conditions reduce the opportunity cost of holding risk assets and tend to support equity valuations. Conversely, the post-pandemic tightening cycle saw real rates move back into positive territory as both inflation rose and the RBI began normalising policy.

While a formal regression of equity returns on real interest rates falls beyond the primary scope of this project, the conceptual importance of real rates as a monetary policy measure is clear: nominal rate changes that simply track inflation movements carry different implications for financial markets than changes that alter the real cost of capital.

6.3 Inflation Regime Analysis

A key contribution of this analysis is the examination of market behaviour across different inflation regimes. Table 5 summarises average NIFTY 50 returns and volatility under each regime.

Table 5: NIFTY 50 Returns by Inflation Regime

Inflation Regime	Threshold	Months	Avg. Monthly Return (%)	Std. Dev. (%)
Normal Inflation	CPI $\leq$ 6%	82	0.83	4.83
High Inflation	CPI $>$ 6%	35	1.70	4.49
Full Sample	All months	119	0.99	4.72

Counter to conventional wisdom, average NIFTY 50 returns were actually higher during high-inflation months (approximately 1.70%) than during normal-inflation months (approximately 0.83%). This seemingly paradoxical result may reflect several factors. First, high-inflation periods in India have occasionally coincided with periods of strong economic growth, where nominal corporate earnings also rise rapidly. Second, commodity-exporting sectors such as energy and materials, which constitute a non-trivial share of the NIFTY 50, tend to benefit directly from the commodity price surges that typically drive CPI higher.

It is important to note that the difference in average returns across regimes is not tested for statistical significance in this preliminary analysis; given the small sub-sample sizes and high return volatility, the observed differential may not be statistically significant. Nevertheless, the directional result is consistent with the broader finding that the relationship between inflation and equity returns in India is not straightforwardly negative, at least at the monthly frequency.

7. Graphical Analysis

The time series and scatter plot visualisations embedded in the accompanying Excel workbook complement the statistical analysis presented above. This section summarises the key visual findings.

7.1 Time Series Plots

The time series charts tracking inflation, repo rate, and NIFTY 50 returns over the 2013–2023 period reveal several important structural features of the data:

- **Inflation Trajectory:** Inflation was persistently high (above 6%) in 2013–2014, before declining sharply to a trough near 1.5% in mid-2017. It then gradually rose and spiked again above 6% in 2022–2023 in response to global commodity price pressures following the Russia-Ukraine conflict.
- **Repo Rate Movements:** The repo rate followed a broadly counter-cyclical pattern: it was raised aggressively in 2013–2014 to combat inflation, then gradually cut from 8% to 4% over 2015–2020 as inflation subsided and growth support became the priority, before being reversed upward from mid-2022 as inflationary pressures resurged.
- **NIFTY 50 Returns:** Monthly returns exhibit no discernible trend, consistent with their transformation from a non-stationary price level to a stationary returns series. Extreme negative returns are visible during specific episodes: the global risk-off period of early 2016, the demonetisation shock of late 2016, and the COVID-19 crash of March 2020. Conversely, exceptional positive months coincide with post-election relief rallies and post-COVID recovery.

7.2 Scatter Plots

The scatter plots of inflation versus returns and repo rate versus returns visually confirm the regression findings. In both plots, the data points are dispersed broadly across the chart with no discernible systematic pattern. The regression line in each plot is nearly flat, reinforcing the conclusion that neither variable exerts a strong directional pull on monthly equity returns.

The absence of outlier-driven patterns is notable: the weak relationship is not attributable to a small number of extreme data points but reflects the broad-based absence of correlation across the full sample.

8. Discussion

8.1 Reconciling Theory and Evidence

The empirical findings of this study appear to contradict the standard theoretical prediction that higher inflation and higher interest rates should be associated with lower equity returns. However, this apparent contradiction can be reconciled through several theoretical and empirical considerations.

First, the standard negative relationship between interest rates and equity valuations is derived from a discounted cash flow (DCF) framework that is most relevant for long-horizon analysis. Over a one-month window, investors do not systematically revalue the entire stream of future corporate earnings in response to a modest repo rate change. High-frequency returns are dominated by sentiment, news flow, and liquidity factors that have little to do with macroeconomic fundamentals.

Second, the Efficient Markets Hypothesis (EMH) provides a direct explanation. In a semi-strong efficient market, all publicly available information—including scheduled inflation data releases and RBI policy announcements—is immediately incorporated into prices. Since both the CPI release and RBI policy decisions are widely anticipated and heavily discussed events, markets likely price in the expected outcome well before the official announcement. The realised macroeconomic data therefore contains little marginal information at the monthly frequency.

Third, the macroeconomic environment is inherently multi-dimensional. Inflation may rise because of demand-side pressures (strong economic growth), which are equity-positive, or supply-side shocks (oil price spikes), which are equity-negative. A simple inflation variable conflates these economically distinct drivers, introducing noise into the estimated relationship.

8.2 Implications for Investors

The practical implication of these findings is that short-term equity market timing based solely on contemporaneous macroeconomic indicators such as CPI inflation or the repo rate is unlikely to yield systematic outperformance. Macroeconomic information is, in a meaningful sense, already in the price by the time it is publicly observable.

This does not imply, however, that macroeconomic analysis is irrelevant for equity investors. Over longer horizons—quarters or years rather than months—the cumulative effect of sustained changes in the monetary policy stance and inflation environment may be significantly more important for portfolio construction and sector allocation decisions. Furthermore, the real interest rate framework suggests that investors should focus not merely on the level of the repo rate but on its relationship to prevailing inflation.

8.3 Limitations

Several limitations of this study merit acknowledgment. First, the analysis is confined to a linear OLS framework, which by construction cannot detect non-linear or threshold effects in the data. Future research could employ threshold regression or Markov-switching models to capture regime-dependent dynamics more rigorously. Second, the model omits potentially important control variables including global equity market conditions (e.g., S&P 500 returns), foreign institutional investor flows, exchange rate movements, and domestic economic growth proxies. Third, the sample period, while covering a rich macroeconomic cycle, is limited to a single decade and encompasses the extraordinary disruption of the COVID-19 pandemic, which may have introduced structural breaks not explicitly accounted for. Finally, the use of monthly data aggregates information across multiple trading days, potentially masking higher-frequency relationships between intraday or weekly market responses to specific policy announcements.

9. Conclusions

This project has examined the relationship between CPI-based inflation, the RBI Repo Rate, and monthly NIFTY 50 equity returns in India over the period January 2014 to November 2023, a sample of 119 monthly observations.

The principal finding is that neither contemporaneous inflation nor the repo rate exerts a statistically significant influence on monthly NIFTY 50 returns within the OLS regression framework. The model's R-squared is negligibly small at 0.54%, and both the F-test for joint significance and the individual t-tests for each regressor fail to reject the null hypothesis of no relationship at conventional significance levels.

These results are interpreted as consistent with the Efficient Markets Hypothesis: the information content of publicly observable, schedule-released macroeconomic data is rapidly incorporated into equity prices, leaving little residual predictive power at the monthly frequency. Additional analyses—including lagged inflation, real interest rates, and inflation regime classification—provide supplementary insights but do not overturn the central conclusion.

The inflation regime analysis yields a counter-intuitive result: average NIFTY 50 returns have been somewhat higher during high-inflation months than normal-inflation months in this sample, a finding that likely reflects the confounding influence of economic growth dynamics and sectoral composition rather than a causal positive effect of inflation on equity returns.

Overall, this research reinforces the view that short-term equity return prediction from macroeconomic variables alone is a challenging task, even in an emerging market context where macroeconomic

volatility is relatively high. Long-horizon analysis, multi-factor modelling, and attention to market structure and expectations formation mechanisms offer more promising avenues for understanding the macroeconomic determinants of Indian equity market performance.

References

Fama, E. F. (1970). Efficient Capital Markets: A Review of Theory and Empirical Work. *The Journal of Finance*, 25(2), 383–417.

Fama, E. F., & Schwert, G. W. (1977). Asset Returns and Inflation. *Journal of Financial Economics*, 5(2), 115–146.

Reserve Bank of India. (2016). Monetary Policy Framework Agreement and Flexible Inflation Targeting. RBI Publications, Mumbai.

Ministry of Statistics and Programme Implementation. (2023). Consumer Price Index – Combined (Rural + Urban). Government of India, New Delhi.

National Stock Exchange of India. (2023). NIFTY 50 Historical Data. NSE India, Mumbai.

Naik, P. K., & Padhi, P. (2012). The Impact of Macroeconomic Fundamentals on Stock Prices Revisited: Evidence from Indian Data. *Eurasian Journal of Business and Economics*, 5(10), 25–44.

Tripathy, N. (2011). Causal Relationship between Macro-Economic Indicators and Stock Market in India. *Asian Journal of Finance & Accounting*, 3(1), 208–226.

Ross, S. A. (1976). The Arbitrage Theory of Capital Asset Pricing. *Journal of Economic Theory*, 13(3), 341–360.